

7

One-Way Between-Subjects ANOVA

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To be frank, the t -test should have died out many years ago. The reason I say so is that the t -test has a very serious limitation: it can only test for the difference between two means. Because of this serious limitation, Ronald Fisher, way back in 1935, introduced a statistic known as the analysis of variance (or ANOVA for short). The ANOVA can handle hypotheses relevant to the difference between two or more means. Consequently, the ANOVA is the best of both worlds. That is, if your analysis has more than two means, the ANOVA will work. Additionally, if your analysis is restricted to the difference between two means, the ANOVA will, again, work fine. So, why not always use ANOVA? Good question. I think we should. However, I do not foresee the popularity of the t -test dying out in my lifetime. Consequently, I felt compelled to include a chapter on the t -test in this textbook. Additionally, I would recommend that you employ a t -test to test hypotheses relevant to the difference between two means, and to employ an ANOVA to test hypotheses relevant to the difference between three or more means. By doing so, you'll reduce the chances of confusing your reader who likely does not know about the big secret (i.e., the t -test should be dead!).

Analysis of Variance: Three Groups – Variances Equal

Based on the available evidence, whether you are a male or a female likely has little influence on your overall cognitive ability (see chapter on t -test). However, whether you are under the influence of alcohol likely does - and the effect is not positive (surprise). Peterson, Rothfleisch, Zelazo, and Pihl (1990) randomly assigned adult participants to three groups: low dose alcohol, medium dose alcohol, and large dose alcohol. The alcohol was served with grapefruit juice to mask the taste/strength of the alcohol (i.e., so participants wouldn't know what group they were in). The participants then completed a battery of cognitive ability tests, including a long-term visual memory test called the Rey-Osterreith. I have simulated data to correspond exactly to the results reported by Peterson et al. (1990). The histograms are reported in Figure C7.1, and the descriptive statistics are reported in the SPSS table entitled 'Statistics'. ([Data File: alcohol memory](#)) ([Watch Video 7.1: Separate Descriptives & Histograms for ANOVA](#))

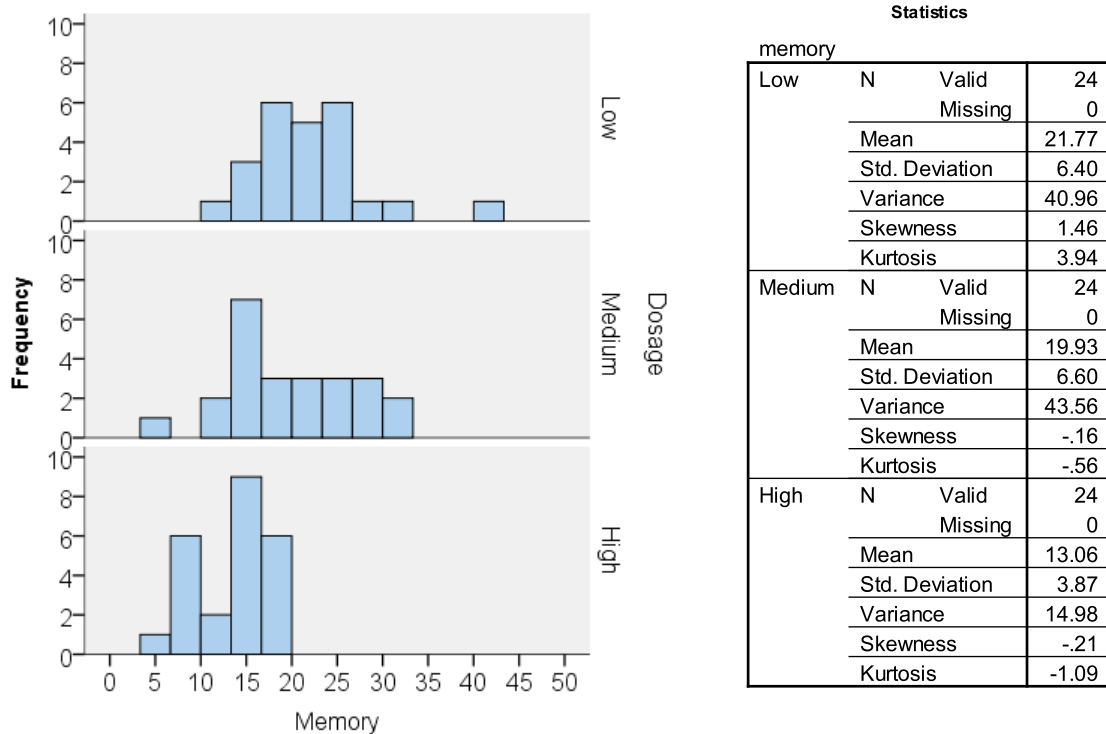
At least numerically, it looks like alcohol had a negative impact on long-term visual memory performance, as the mean performance on the long-term visual memory test was 21.77 for the low-dose group and 13.06 for the high-dose group. The medium-dose group scored somewhere in the middle at 19.93. Of course, the statistical question is, did these numerical

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mean differences arise simply due to chance? Such a statistical question can be answered with what is known as the one-way between-subjects ANOVA (a.k.a., one-way between-.

I mention here only briefly that the assumptions associated with the one-way between subjects ANOVA are identical to the assumptions associated with the independent samples t -test: random sampling, independence of observations, dependent variable measured on an interval/ratio scale, normally distributed data, and homogeneity of variance. Additionally, the robustness of the t -test to violations of the dependent variables measured on an interval/ratio scale, normality, and homogeneity of variance assumptions described in the t -test chapter extend to the one-way between subjects ANOVA.

Figure C7.1. Histograms and Descriptive Statistics for Alcohol Dosage and Memory Performance



Hypotheses

Null hypothesis (H_0): The memory means will be equal across all three alcohol dose groups

Alternative hypothesis (H_1): The memory means will not be equal across all three alcohol dose groups

One-way Between-Subjects ANOVA: Three Groups – Formulae

In this section, I will describe the nature of ANOVA with some accessible formulae. If you're not interested in understanding the nature of the between-subjects ANOVA, feel free to skip this section and jump to the section entitled 'One-Way Between-Subjects ANOVA: SPSS'.

You may be surprised to learn that, when we are interested in testing the difference between means statistically, we use something called ‘analysis of variance’. Shouldn’t it be called ‘analysis of means’, instead, I hear you cry? The answer is: ‘No, it shouldn’t.’ To understand why, you need to have some understanding of the fundamental nature of ANOVA.

As per the independent samples *t*-test, ANOVA is a ratio, or an *F*-ratio more precisely, in deference to its inventor, Ronald Fisher. Again, as per the independent samples *t*-test, ANOVA is the ratio of the “good stuff” to the “bad stuff”. The “good stuff” is the variance associated with the effect of interest. In this example, the variability in memory performance due to alcohol consumption is the “good stuff”. The “bad stuff” is the variance associated with the effect of interest *plus* sources of variance that we don’t really have any idea about. Essentially, the only difference between the independent samples *t*-test and the one-way between-subjects ANOVA is that the latter can accommodate hypotheses that include more than two means/groups. In order to perform the calculations involved with a between-subjects ANOVA, two sources of variance need to be estimated: between-subjects and within-subjects.

In the independent samples *t*-test case, the variability in the two means was represented simply by the range (one mean minus the other mean). When you have three means or more, you need a more sophisticated measure of variability. For example, the following formula can be used to help calculate the variability among several group means. It is known as the Sums of Squares Between (SS_B):

$$SS_B = N \sum (\bar{X}_j - \bar{X}_G)^2 \quad (1)$$

where N = sample size per group (this formula assumes equal sample sizes), $\bar{X}_j$ is equal to the group means of interest, and $\bar{X}_G$ is equal to the grand mean. The grand mean is the average of the means of interest (in this example $\bar{X}_G = 18.253$). Thus,

$$SS_B = 24 * ((21.77 - 18.253)^2 + (19.93 - 18.253)^2 + (13.06 - 18.253)^2)$$

$$SS_B = 24 * (12.369 + 2.812 + 26.967)$$

$$SS_B = 24 * 42.148$$

$$SS_B = 1011.552$$

It can be appreciated above that there were numerical differences between the three group means and the grand mean. As the squared differences increase, the chances of rejecting the null hypothesis increase, all other things equal. I used the formula above which assumes equal sample sizes across groups for the sake of simplicity (in this example $N = 24$). ANOVA can handle unequal sample sizes with a slight variation to the above SS_B formula, which we won’t worry about, for now.

The “bad stuff” in the context of the ANOVA *F*-ratio can be estimated with something known as Sums of Squares Within (SS_W):

$$SS_W = \sum (N_j - 1)S_j^2 \quad (2)$$

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where N is equal to the sample size of each group (i.e., 24) and S_j^2 is equal to the variance associated with each group. Thus, based on the variances reported in the SPSS table entitled 'Statistics' above:

$$SS_W = ((24 - 1) * 40.96) + ((24 - 1) * 43.56) + ((24 - 1) * 14.98)$$

$$SS_W = 942.08 + 1001.88 + 344.54$$

$$SS_W = 2288.50$$

Thus, as the variances associated with each group increase, the chances of rejecting the null hypothesis decrease, all other things equal. For completeness, one can also calculate something known as the Sums of Squares Total, which is simply $SS_B + SS_W$.

$$SS_T = SS_B + SS_W \quad (3)$$

Therefore, $SS_T = 3300.052$

$$SS_T = 1011.552 + 2288.50$$

$$SS_T = 3300.052$$

As I demonstrate below, SS_T comes into play for the purposes of calculating an effect size estimate for ANOVA (known as eta-squared or η^2).

Degrees of Freedom

Although the important sources of variance associated with the ANOVA have been calculated (SS_B and SS_W), they are not sufficient, on their own, to calculate the F -ratio. Instead, the sums of squares need to be averaged, because they have to be placed on a comparable metric. Consider that only three values were used to calculate the SS_B (i.e., the three means), whereas the variances associated with the SS_W were multiplied by their respective sample sizes (i.e., 24). It should come as no surprise that SS_W is numerically larger than SS_B . Consequently, it doesn't make sense to calculate the ANOVA F -ratio based on SS_B/SS_W . To place SS_B and SS_W on a comparable metric, they need to be averaged, and this is done by dividing the SS_B and SS_W by their respective degrees of freedom.

In the one-way between-subjects ANOVA, the df associated with SS_B is equal to $k-1$, where k is equal to the number of groups (or means) in the analysis. In this example, the df associated with SS_B is $3-1 = 2$. In the one-way between-subjects ANOVA, the df associated with the SS_W is equal to $N - k$. Thus, in this example, the df associated with the SS_W is equal to $72 - 3 = 69$. When SS_B and SS_W are divided by their respective degrees of freedom, the result is known as 'Mean Squares' (MS). Thus, the MS_B , in this example, is equal to $1011.55/2 = 505.78$. By contrast, the MS_W , in this example, is equal to $2288.50 / 69 = 33.17$.

In computational terms, The F -ratio in the ANOVA is the ratio of MS_B (the good stuff) and the MS_W (the bad stuff). Therefore, in this example, $F = 505.78 / 33.17 = 15.25$. I have placed all of the important values associated with the calculation of an F -ratio in the Table C7.1. Such a presentation of ANOVA related calculations is known as an 'ANOVA summary table'. You can

see that there is one value that remains to be calculated: the p -value. The p -value will determine whether the null hypothesis can be rejected or not. More specifically, the p -value will address the following question, 'Is the F -value of 15.25 large enough to be considered statistically significant ($p < .05$)?'

Table C7.1. ANOVA Summary Table

Source	SS	df	MS	F	p
Between	1011.55	2	505.78	15.25	?
Within	2288.50	69	33.17		
Total	3300.05	71			

Note. SS = Sums of Squares; df = degrees of freedom; MS = Mean Squares; F = F-ratio; p = probability.

Determining the Statistical Significance of an F -value

Fisher (1935) demonstrated that the F -ratio follows (wait for it) the F -distribution, when the null hypothesis is true. Typically, researchers “hope” to reject the null hypothesis. Consequently, they hope the F -ratio they obtained will be larger than what is expected by chance sampling. As a general statement, an F -ratio of approximately 1.0 has no real chance to reject the null hypothesis, as it would imply that the MS_B is no larger than the MS_W . However, as the F -ratio exceeds 1.0 in magnitude, the chances that the null hypothesis will be rejected increase. Note that, unlike the t -values, it is not possible to obtain a negative F -value. All F -values are positive. An F -value can be less than 1.0 due to sampling fluctuations, but not less than 0.

Fisher (1935) created an F -distribution table with which to compare calculated F -values, based on specified degrees of freedom. In the olden days (pre 2000), that is what researchers did to determine whether a between-subjects ANOVA was statistically significant. Today, people use computers to estimate the p -value for a statistical effect. SPSS can calculate a precise p -value for an obtained F -value through the following SPSS compute statement (**Watch Video 7.2: Compute p-value for F-value in SPSS**):

Transform -> Compute-> Sig.F

Based on an obtained F -value of 15.25, and degrees of freedom of 2 and 69, the p -value was estimated at .000003276, which is much less than .05. Consequently, the null hypothesis of equal long-term visual memory means across the three alcohol consumption groups can be rejected, $p < .001$.

Unfortunately, the statistically significant F -value does not tell us which of the three means are statistically significantly different from each other. It only tells us that there is at least one comparison that is statistically significant amongst the three means. For this reason, ANOVA is known as an **omnibus statistic**. Almost certainly, the difference between the low-dose group and the high-dose group means will be found to be statistically significant, because they are the

most substantially different numerically. However, we do not know if the low-dose group and the medium-dose group means are statistically significantly different from each other; nor do we know if the medium-dose and high-dose group means are statistically significantly different from each other. Again, the ANOVA is only telling us that at least one comparison is statistically significant, and it is up to us to find out which one(s) is (are). A simple option would be to follow-up the statistically significant ANOVA with a series of independent samples *t*-tests. Prior to considering that option, however, I would like to demonstrate an efficient option in SPSS to calculate an ANOVA summary table, including the *p*-value, simultaneously. Then, I will cover the topic of effect size in the one-way between-subjects case. Finally, I will discuss multiple comparison procedures.

One-Way Between-Subjects ANOVA: SPSS

In order to conduct a one-way between-subjects ANOVA in SPSS via the menus, there are two options. In this chapter, I will only demonstrate the option, via the One-way ANOVA utility, which includes the simpler output to interpret (**Watch Video 7.3: [Oneway Between-Subjects ANOVA in SPSS](#)**).¹

ANOVA					
memory					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	1011.573	2	505.786	15.250	.000
Within Groups	2288.429	69	33.166		
Total	3300.002	71			

As can be seen in the table entitled 'ANOVA', the *F*-value was estimated at 15.25 and the *p*-value was estimated at .000. Consequently, as .000 is less than .05, the null hypothesis of equal means across all three groups can be rejected. I'll note that researchers are advised to report a *p*-value of .000 as $p < .001$, because there are no certainties in probability in theory. These results are identical to the results I obtained "by hand" in the preceding section.

Before proceeding to the next section, the assumption of homogeneity of variances was tested via Levene's *F*-test. The assumption was satisfied, $F(2, 69) = 2.54, p = .086$, as can be seen in the SPSS table entitled, 'Test of Homogeneity of Variances'. Thus, the variances associated with the groups (i.e., 40.96, 43.56, 14.98) were not found to be statistically significantly different from each other.

¹ I demonstrate the option which includes more complicated output in the chapter devoted to the factorial between-subjects ANOVA.

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Test of Homogeneity of Variances

memory			
Levene Statistic	df1	df2	Sig.
2.543	2	69	.086

One-Way Between-Subjects ANOVA Effect Size: Eta-Squared (η^2)

As you may recall, it is not enough to simply state that a null hypothesis has been rejected with a certain level of probability ($p < .05$). Instead, it is always good practice to complement a statistical analysis with an estimate of effect size. An estimate of effect size will help determine whether the statistically significant difference between the means is large or not. In the context of a one-way ANOVA, the most typically reported measure of effect size is known as eta-squared (η^2):

$$\eta^2 = \frac{SS_{Between}}{SS_{Total}} \quad (4)$$

η^2 represents the proportion of variance in the dependent variable that is accounted for by the independent variable. If that reminds you of the coefficient of determination (Pearson correlation), you have made an important connection: r^2 and η^2 are fundamentally similar. In this example, η^2 was equal to .307:

$$\eta^2 = \frac{1011.55}{3300.05} = .307$$

When η^2 is multiplied by 100, one can speak of percentage of variance accounted for in the dependent variable, rather than proportion of variance. Thus, in this example, 30.7% of the variance in long-term visual memory was accounted for by the level of alcohol consumed by the participants (low-dose, medium-dose, and high-dose). It is possible to calculate confidence intervals for eta-squared, as I discuss and demonstrate in the Advanced Topics section of this chapter.

Because SS_B and SS_W are equal to SS_T , η^2 can also be formulated as:

$$\eta^2 = \frac{SS_{Between}}{SS_{Between} + SS_{Within}} \quad (5)$$

This may seem like a rather pedantic observation to point out, however, the fact that SS_W plays a part in the calculation of SS_T complicates the estimation of η^2 in other types of ANOVAs, for example, the within-subjects ANOVA and the factorial ANOVA (as I discuss in other chapters in the context of partial eta-squared) (**Watch Video 7.4: [What is Eta-Squared?](#)**).

ANOVA Effect Size: Omega-Squared (ω^2)

Although η^2 is intuitive to understand based on the first formula, it has been known for a long time that it is not a fully accurate estimate of effect size. Specifically, eta-squared (η^2) has been shown to be an overestimate of the effect size in the population (Kesselman, 1975). An

adjusted version of η^2 was introduced that is not biased. It is known as omega squared (ω^2 ; Hays, 1963) and can be formulated as:

$$\omega^2 = \frac{F - 1}{F + \frac{(N - k + 1)}{k - 1}} \quad (6)$$

Where F is equal to the ANOVA F -value, N = total sample size, and k is the number of groups included in the analysis. Therefore, based on the F -value included in the ANOVA summary in Table C7.1, ω^2 was estimated at .284:

$$\omega^2 = \frac{15.25 - 1}{15.25 + \frac{(72 - 3 + 1)}{3 - 1}}$$

$$\omega^2 = .284$$

Thus, based on ω^2 , which is a non-biased estimate of effect size, it may be said that 28.4% of the variance in long-term visual memory was accounted for by the level of alcohol consumed by the participants (low-dose, medium-dose, and high-dose). Curiously, despite the fact that nobody would argue that η^2 is more accurate than ω^2 , the vast majority of researchers report η^2 , not ω^2 . In fact, according to google scholar, in 2014, the term ‘eta-squared’ appeared 4,910 times in academic papers. By contrast, omega-squared appeared a measly 212 times. My best guess is that people report η^2 because programs such as SPSS do not report ω^2 ; they report only η^2 .² Another reason ω^2 has not become popular is that it is relatively complicated to calculate ω^2 for factorial ANOVA designs (see Olejnik & Algina, 2003). My impression is that there is not that much difference between η^2 and ω^2 when the sample size in each group is greater than about 30 and the effect size is less than .75. However, when the sample size is less than 20 in each group, it is definitely recommended to calculate ω^2 as an indicator of effect size in the one-way between-subjects ANOVA.

It is possible to calculate confidence intervals for η^2 and ω^2 , however, I am not familiar with any easily implemented procedures for doing so. Personally, I do not think it is worth the effort to calculate confidence intervals for omnibus indicators of effect size, such as η^2 and ω^2 , because they tend to amalgamate multiple sources of effects. For example, when three or more groups are analyzed together, as is typically the case with an ANOVA, it is entirely possible that the entirety of the η^2 or ω^2 may be due to a single mean comparison, rather than distributed across all of the mean comparisons. In almost all cases, the statistically significant ANOVA simply tells us that we have more work to do. Consequently, interpreting η^2 or ω^2 is not really that meaningful, in my opinion. Nonetheless, it should be reported.

² SPSS provides the option to calculate eta-squared in the ‘General Linear Model’ menu option. This procedure is described in the factorial between-subjects ANOVA chapter.

Effect Size Guidelines: Eta-Squared and Omega Squared

Finally, Cohen (1988) provided guidelines for interpreting η^2 from a more practical perspective (see Table C7.2). It can be seen that a large η^2 corresponds to a value of .138. Thus, in the alcohol and memory study, the effect size of .307 would be considered large, based on Cohen's (1988) guidelines. Cohen's (1988) guidelines for interpreting η^2 are probably just as appropriately applied to ω^2 .

Table C7.2. Cohen η^2 Guidelines

Small	Medium	Large
.01	.059	.138

Multiple Comparisons and Type I Error Rate Inflation

As mentioned above, a statistically significant one-way between-subjects ANOVA tells you that at least one mean comparison is statistically significant, but you don't know which one(s). Researchers typically follow-up a statistically significant ANOVA with a series of tests known as 'multiple-comparisons'³ to uncover the statistically significant comparisons. In the context of the one-way between-subjects ANOVA, multiple-comparison tests are essentially a series of modified independent samples *t*-tests (or very similar *q*-tests which follow the *q*-distribution). However, in most cases, the procedure of simply following-up a significant ANOVA with a series of independent samples *t*-tests is not recommended for several reasons. The most important reason relates to the fact that, as more and more analyses are performed on the same sample of data, the chances of committing a type I error (type one error) increases.

Recall that researchers specify the alpha (α) level for each statistical analysis they conduct (usually, $\alpha = .05$). The alpha level corresponds to the maximum level of chance one is willing to accept with respect to rejecting the null hypothesis erroneously. Stated alternatively, the alpha level corresponds to the chances that a researcher will claim that there is a statistical effect, when, in fact, there is no effect in the population. In statistics, deciding that there is a statistically significant effect, when in fact there is no statistically significant effect, is known as a type I error. As I will describe further below, researchers very commonly commit type I errors. In fact, it has been argued that many apparent discoveries published in science journals are actually type I errors (Ioannidis, 2005; Simmons, Nelson & Simonsohn, 2011). Consequently, it is very important to help protect oneself from committing type I errors.

In the context of testing the difference between two or more means, another term that researchers use, instead of alpha level, is 'per contrast error rate'. Typically, researchers specify the per contrast error rate at .05, which is the same as alpha. Occasionally, when a researcher

³ Many researchers use the term 'post-hoc tests' in this context, but I prefer the term 'multiple-comparisons'.

wants to be conservative, perhaps because the implications of the results are very serious, the per contrast error rate (or alpha) may be specified at .01. In those cases, the p -value calculated for a particular statistic (say, t -test) has to be lower than .01, in order to be statistically significant. Ultimately, the researcher specifies the alpha level/per contrast error rate. Usually, it is .05; that is, a 5% chance of rejecting erroneously the null hypothesis when it is in fact true.

One of the challenges facing the typical researcher is that for each statistical analysis he or she performs on the *same* sample of data, the chances of committing at least one type I error increases. That is, it increases beyond .05. The chances of committing one or more type I errors across a collection of statistical analyses performed on the same sample of data is known as the **familywise error rate**. The familywise error rate associated with a collection of statistical tests is essentially additive, up to a point. For example, if a researcher performed three different independent sample t -tests on the same sample of data with alpha specified at .05, each individual t -test would be associated with a per-contrast error rate of .05. However, as a collection of statistical analyses, the chances of committing at least one type I error would be equal to approximately .15 (i.e., $.05 * 3$).

I wrote ‘approximately’, because it is not simply the case that the per-contrast error rate needs to be multiplied by the number of comparisons. To obtain a more accurate estimate of the familywise error rate, the following ‘binomial formula’ should be used (**Watch Video 7.5: [What is familywise error rate?](#)**):

$$\alpha_{FW} = 1 - (1 - \alpha_{PC})^c \quad (7)$$

where c is equal to the number of comparisons performed and α_{PC} is equal to the specified per contrast error rate (i.e., alpha, usually .05). Thus, in the case of three comparisons, the α_{FW} error rate would be expected to equal:

$$\begin{aligned} \alpha_{FW} &= 1 - (1 - .05)^3 \\ &= 1 - (.95)^3 \\ &= 1 - (.857) \\ &= .143 \end{aligned}$$

The more accurate estimate of .143 is slightly smaller than the previous “quick and dirty” estimate of .15. Because researchers want to avoid embarrassing themselves by committing a type I error, they are encouraged to employ statistical techniques to help protect themselves from doing so. Unfortunately, researchers do not always do so. Next, I describe the distinction between two broad types of multiple comparisons procedures that can help keep the familywise error rate close to .05.

Single-Step Versus Two-Step Multiple Comparison Procedures

A very large number of methods (30+) have been proposed to control the familywise error rate to approximately .05. The degree to which they do so has been the subject of a

substantial amount of research. Some methods are excessively conservative, while others are excessively liberal. In practice, the challenge is to find a method that can keep the familywise error rate close to .05, while also relatively easy to implement. The most well-known techniques fall into two broad categories: (1) two-step multiple comparison procedures, and (2) single-step multiple comparison procedures.

Two-step procedures require a statistically significant ANOVA to be observed, prior to conducting the follow-up multiple comparisons (e.g., Fisher's LSD; Fisher-Hayter HSD). By contrast, single-step procedures do not require a statistically significant ANOVA to be observed (e.g., Tukey-Kramer HSD; Games-Howell). Thus, with respect to single-step procedures, researchers should go straight to the multiple comparison analyses and ignore the ANOVA results.

In my opinion, most researchers should be able to get on with life satisfactorily knowing only five multiple-comparison procedures. I have listed them in Table C7.3. It can be seen that they are all associated with one or more virtues. Consequently, most researcher scenarios should be covered by the five multiple comparison procedures listed in Table C7.3. First, I describe Fisher's protected LSD (least significant difference), because it is the most attractive option to apply in the alcohol and memory study. It is also the simplest to understand.

Table C7.3. Five Key Multiple Comparisons Procedure and their Characteristics

	Steps	Means	Variances	Distribution	Non-Pairwise	CIs
Fisher's LSD	Two-step	3	Equal	t	No	No
Tukey-Kramer's	Single-step	3+	Equal	q	No	Yes
Fisher-Hayter	Two-step	3+	Equal	q	No	No
Bonferonni	Single-step	3+	Equal	t	Yes	Yes
Games-Howell	Single-step	3+	Unequal	q	No	Yes

Note. CIs = Confidence intervals

Fisher's Protected Least Significant Difference (LSD)

It should be mentioned, first, that the protected Fisher's LSD has a poor reputation in the research community. This reputation is not justified, or, at least, not justified fully. Specifically, many sources state that the protected Fisher's LSD does not keep the familywise error rate at alpha .05. However, both analytical and simulation research has shown convincingly that when an analysis includes only three groups, as in the alcohol and memory example (low-dose, medium-dose, and high-dose), the protected Fisher's LSD does keep the familywise error rate at .05. For example, Hayter (1986) demonstrated analytically that with three groups, the protected Fisher's LSD maintains the familywise error rate at .05. However, as soon as you move

up to four groups, the protected Fisher's LSD per-contrast error rate blows out to .122, so it is not useful in such a scenario. Additionally, Seaman, Levin & Serlin (1991) demonstrated via simulation that the protected Fisher's LSD keeps the familywise error rate at .05, when the analysis is based on three means. Furthermore, depending on how one defines statistical power, the protected Fisher's LSD was found to be somewhere between 8% and 25% more powerful than the commonly used Tukey's HSD multiple-comparison procedure (Seaman et al., 1991). Although it may seem like the analysis of three groups is a rather narrow application of the protected Fisher's LSD, in practice, a very large percentage of hypotheses include only three means. In fact, it is probably the most commonly observed one-way between-subjects ANOVA in the literature, and, yet, very few people use the protected Fisher's LSD in this context.

The protected Fisher's LSD is known as 'protected', because it must be preceded by a statistically significant ANOVA. The follow-up Fisher's LSD tests are very similar to independent *t*-tests. The only mathematical difference between the Fisher's LSD tests and independent samples *t*-tests is that Fisher's LSD uses a pooled estimate of variance for the purposes of estimating the standard error of the difference between the means. Stated alternatively, all of the mean comparisons in a family use the same standard error of the difference between means. Here is the formula for Fisher's LSD:

$$\text{Fisher's LSD} = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{MS_w \left(\frac{1}{N_1} + \frac{1}{N_2} \right)}} \quad (8)$$

You can see that the numerator is the difference between two means (let's say low-dose alcohol versus medium-dose alcohol). The denominator is the MS_w estimated from the one-way ANOVA multiplied by a sample size factor (square rooted). Because MS_w is based on all of the group variances included in the one-way ANOVA, it is known as a pooled estimate. By contrast, the independent samples *t*-test denominator includes only the two variances associated with the two means included in the numerator. Consequently, Fisher's LSD and the independent samples *t*-test will tend to provide slightly different *t*-values, in most cases.

Fisher's LSD: Example

Recall that the one-way between-subjects ANOVA for the alcohol and memory performance analysis was statistically significant, $F(2, 69) = 15.25$, $p < .001$. Furthermore, the analysis included only three groups. Thus, it would be appropriate to follow-up the analysis with a series of Fisher's LSD tests. With respect to the low-dose alcohol and medium-dose alcohol Fisher's LSD comparison, the formula and solution would look like this:

$$\text{Fisher's LSD} = \frac{21.77 - 19.93}{\sqrt{33.17 \left(\frac{1}{24} + \frac{1}{24} \right)}}$$

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$$\text{Fisher's LSD} = \frac{1.84}{\sqrt{33.17(.042 + .042)}} = \frac{1.84}{1.669} = 1.10$$

If a t -test value of 1.10 seems small to you, you would be right to think so. As mentioned above, Fisher's LSD is essentially a t -test, and it follows the t -distribution. Consequently, the Fisher's LSD t -value of 1.10 can be tested for statistical significance against the t -distribution. In this case, the analysis has 69 degrees of freedom ($df = N - k$; thus, $df = 72 - 3 = 69$).

It is possible to estimate a p -value for any t -value in SPSS via the Transform -> Compute Variable utility⁴, however, it is more efficient to get SPSS to conduct a series of protected Fisher's LSD tests automatically. To do so, you can use the Post-Hoc menu option in SPSS (**Watch Video 7.6: Protected Fisher's LSD in SPSS**). In addition to re-running the one-way between-subjects ANOVA, SPSS will produce the following additional table entitled 'Multiple Comparisons'.

Multiple Comparisons

Dependent Variable: memory
LSD

(I) dosage	(J) dosage	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Low	Medium	1.84000	1.66247	.2722295	-1.4765	5.1565
	High	8.71000*	1.66247	.0000017	5.3935	12.0265
Medium	Low	-1.84000	1.66247	.2722295	-5.1565	1.4765
	High	6.87000*	1.66247	.0000993	3.5535	10.1865
High	Low	-8.71000*	1.66247	.0000017	-12.0265	-5.3935
	Medium	-6.87000*	1.66247	.0000993	-10.1865	-3.5535

*. The mean difference is significant at the 0.05 level.

SPSS will typically report p -values to three decimal places. The number of decimal places can be changed easily, if necessary (**Watch Video 7.7: Increase Decimal Place in SPSS output**). In the first row of the SPSS 'Multiple Comparisons' table, you will see the comparison between low-dose and medium-dose groups was associated with a mean difference of 1.84 and a standard error of the difference between means equal to 1.662. Those values are nearly identical to the values calculated with the Fisher's LSD formula above (i.e., 1.669; difference due to rounding). It can also be seen in the SPSS 'Multiple Comparisons' table that a p -value of .272 was estimated for the low-dose and medium-dose mean comparison. Consequently, as the p -value was not less than .05, the null hypothesis of equal means was not rejected.

In contrast to the low-dose and medium-dose mean comparison, the mean difference ($\Delta M = 8.71$) between the low-dose ($M = 21.77$) and the high-dose ($M = 13.06$) groups was statistically significant, $p < .001$. Finally, the mean difference between medium-dose and high-dose ($\Delta M = 6.87$) was also found to be statistically significant, $p < .001$. Unfortunately, and very

⁴ The t -value must be squared and then use the F -distribution ($t^2 = F$)

curiously, SPSS does not report the t -values associated with the Fisher's LSD tests. However, the t -values can be calculated by dividing the mean difference by the standard error of the difference between the means.

It should be emphasized that the Fisher's LSD test demonstrated in this chapter is known as the "protected Fisher's LSD", because a one-way between-subjects ANOVA was conducted first and found to be statistically significant. That is, the three Fisher's LSD comparisons were protected from inflating the familywise error beyond .05, given the statistically significant ANOVA. Had the ANOVA not been statistically significant, it would not have been appropriate to proceed with a series of three Fisher's LSD comparisons.

Fisher's LSD and Cohen's d

Although Cohen's d could be calculated for each of the three comparisons with the regular Cohen's d formula, strictly speaking, doing so would not be completely accurate. The reason is that Fisher's LSD is based on a pooled error term across all three groups. If the assumption of homogeneity of variance has been satisfied (discussed further below), such a pooled error term would be a more accurate estimate, in comparison to any combinations of two group variances (or SDs). Stated alternatively, a multiple comparison procedure such as Fisher's LSD is conducted in such a way that the group variances are averaged across groups. Consequently, the corresponding Cohen's d values across all of the comparisons should be based on an averaged error term. The following Cohen's d formula is consistent with the pooled error term principle, as it is based on the Fisher's LSD t -values (i.e., which are calculated based on a common standard error of the difference between means):

$$d = t^* \sqrt{\frac{1}{N_1} + \frac{1}{N_2}} \quad (9)$$

The above Cohen's d formula can handle equal and unequal sample sizes. Although Fisher's LSD is a t -test, unfortunately, SPSS does not report the t -values associated with the Fisher's LSD tests, as noted above. Fortunately, the t -values can be calculated by dividing the difference between the two means by the standard error of the difference between the means (**Watch Video 7.8:** [How to obtain Fisher's LSD \$t\$ -values in SPSS](#)).

$$\text{Fisher's LSD } t = \frac{\text{Mean Difference}}{\text{Std. Error}} \quad (10)$$

As can be seen in Table C7.4, the t -values ranged in value from 1.11 to 5.25. I have also reported the p -values in the table, because I will apply something known as a Bonferonni correction later in the chapter.⁵ It can be observed that two of the three mean comparisons were statistically significant ($p < .05$). Finally, it can be seen that the Cohen's d values ranged

⁵Typically, I would not report p -values beyond three decimal places. However, in order to demonstrate a Bonferonni correction (demonstrated below), one needs non-zero p -values. Consequently, I expanded the number of decimal places to seven.

from .32 to 1.52. Based on Cohen's (1988) guidelines, the low vs. medium d -value would be considered small, whereas the other two d -values would be considered large. Note that it is arbitrary whether the Cohen's d values are reported as positive or negative, as it depends on which mean you decided to subtract from the other.

Table C7.4. Fisher's LSD t -value Calculations and Corresponding Cohen's d Calculations

Low vs. Medium	Low vs. High	Medium vs. High
$t = \frac{1.84}{1.66} = 1.11$	$t = \frac{8.71}{1.66} = 5.25$	$t = \frac{6.87}{1.66} = 4.14$
$p = .2722295$	$p = .0000017$	$p = .0000993$
$d = 1.11 * \sqrt{\frac{1}{24} + \frac{1}{24}} = .32$	$d = 5.25 * \sqrt{\frac{1}{24} + \frac{1}{24}} = 1.52$	$d = 4.14 * \sqrt{\frac{1}{24} + \frac{1}{24}} = 1.20$

Bonferroni Correction

The Bonferroni correction is one of the oldest multiple comparison procedures that has been suggested to keep the familywise error rate at approximately .05 or less. It is a single-step procedure (thus, no need to conduct an ANOVA first). Typically, the Bonferroni correction consists of dividing the specified alpha level by the number of statistical analyses to be performed. Then, the statistical analyses are performed and the obtained p -values from those analyses are compared against the adjusted critical alpha level. For example, if three statistical analyses were to be performed on a sample of data, and a researcher wished to keep the familywise error rate at .05, the researcher would have to specify alpha at .017:

$$\text{Bonferroni adjusted } \alpha = .05 / 3 = .017$$

The Bonferroni correction is well-known and easy to implement. However, it has been found to be too conservative in most cases, particularly as the number of comparisons increase (Ury, 1976). For example, based on five multiple comparisons ($N = 15$), Seaman et al. (1991) found that the Bonferroni adjustment kept the familywise error rate at somewhere between .018 and .037, rather than .05.

In the alcohol and memory example, there were three multiple comparisons. Consequently, the per contrast error rate needs to be adjusted to .017, in order to keep the familywise error rate at .05, using the conventional Bonferroni approach. Rather than dividing the per-contrast error rate by the number of comparisons, an alternative approach is to multiply the unadjusted p -values by the number of comparisons that were performed. Those adjusted p -values that are less than .05 would be deemed significant statistically, $p < .05$, while controlling the familywise error rate at .05.

In the current alcohol and memory study example, the two statistically significant comparisons remained significant statistically, after Bonferroni adjustment, as can be seen in

Table C7.5. In practice, the effects of the application of Bonferroni adjustment will be to observe several multiple comparisons “lose” their statistical significance (**Watch Video 7.9: [What is the Bonferroni correction? Applied in SPSS](#)**).

Table C7.5. *Fisher’s LSD unadjusted and Bonferroni adjusted p-values*

	Low vs. Medium	Low vs. High	Medium vs. High
Unadjusted	$p = .2722295$	$p = .0000017$	$p = .0000993$
Adjusted	$p = .8166885$	$p = .0000051$	$p = .0002979$

Note. Adjusted = unadjusted p -value multiplied by c ; c = number of comparisons; in this example, $c = 3$.

Personally, I would avoid using the Bonferroni adjustment, given that there are several much more attractive alternatives. However, the Bonferroni adjustment is commonly applied, consequently, it is important to have some knowledge of the procedure. Additionally, the Bonferroni adjustment can be applied when testing non-pairwise comparison, as discussed in the Advanced Topics section of this chapter.

Tukey-Kramer’s Honestly Significant Difference (HSD)

Perhaps the most popular single-step multiple comparison procedure is Tukey-Kramer’s HSD.⁶ Bizarrely, Tukey specifically developed the statistic as a single-step procedure, but researchers the world over first conduct an ANOVA, and then follow-up with the Tukey-Kramer’s HSD mean comparisons. The ANOVA is fundamentally unnecessary, in this case.

Tukey-Kramer’s HSD is essentially a modified independent samples t -test, although it uses a statistical distribution slightly different to the t -test. The statistical distribution is known as the q distribution. Simulation research has shown that Tukey-Kramer’s HSD to be fairly conservative (Hayter, 1984). Specifically, it tends to keep the familywise error rate closer to .025, rather than .05 (Seaman et al., 1991). For this reason, I prefer not to use it, especially when the number of groups is only three. However, because Tukey-Kramer’s HSD is so popular, I felt compelled to demonstrate the procedure. Also, Tukey-Kramer’s HSD can provide simultaneous confidence intervals, which is an attractive feature, because two-step multiple comparison procedures do not offer that opportunity. Finally, Tukey-Kramer’s HSD can handle four or more

⁶ Many researchers refer to Tukey-Kramer’s HSD as simply Tukey’s HSD, however, this is not appropriate, because Tukey’s HSD assumes equal sample sizes, whereas Tukey-Kramer’s HSD does not. In practice, there is no point to ever apply Tukey’s HSD. Most statistical programs apply Tukey-Kramer’s HSD, not Tukey’s HSD, even if they do not specify so (e.g., SPSS).

means in an analysis, whereas the protected Fisher's LSD cannot. Next, I describe the use of Tukey-Kramer's HSD with an example from the literature.

Analysis of Variance: Four Groups or More - Tukey-Kramer's HSD

Are you good at detecting a lie? Do you think you're better than a professional whose job often involves detecting lies? Ekman, O'Sullivan, & Frank (1999) paid actors to lie on camera about various issues. The actors were also paid to tell the truth about various issues. Ekman et al. (1999) chose the especially convincing recordings and presented them (50% lying video recordings; 50% honest video recordings) to various groups of professionals whose job involves, at least to some degree, detecting lies. The four groups of professionals included federal agents who do a lot interrogations (mostly CIA agents), federal judges, police officers, and clinical psychologists ([Data File: catch a liar](#)).

As can be seen in Table C7.6, federal agents achieved a mean accuracy score of 73%. By contrast, police officers were essentially performing at chance (50.8%).

Table C7.6. Descriptive Statistics

	Federal Agents	Federal Judges	Police Officers	Clinical Psychologists
<i>M</i>	73.00	62.00	50.80	62.10
<i>SD</i>	13.60	14.90	17.80	15.20
Skew	-.12	.17	.77	-.15
Kurtosis	-.38	-.29	.41	-.09
<i>N</i>	23	84	36	209

Whether any of the numerical differences were statistically significant remains to be determined. The data were sufficiently normally distributed for the purposes of conducting a parametric statistic such as a between-subjects ANOVA and/or a series of Tukey-Kramer's HSD tests.

As can be seen in the SPSS table entitled 'Test of Homogeneity of Variances', the assumption of homogeneity of variance has also be satisfied via Levene's test: $F(3, 348) = .45$, $p = .715$, therefore, the Tukey-Kramer's HSD results can be interpreted.

Test of Homogeneity of Variances

accuracy

Levene Statistic	df1	df2	Sig.
.453	3	348	.715

Recall that there is no need to conduct an ANOVA, when Tukey-Kramer's HSD is selected as the multiple-comparison procedure. However, for educational purposes, I have reported the ANOVA table below (SPSS produces the table automatically). It will be noted that the null

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hypothesis of equal means in the capacity to detect lying across the four groups was rejected, $F(3, 348) = 10.31, p < .001, \eta^2 = .082$.

ANOVA

accuracy					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	7253.412	3	2417.804	10.306	.000
Within Groups	81641.848	348	234.603		
Total	88895.260	351			

Multiple Comparison: Tukey's HSD in SPSS

To perform Tukey-Kramer's HSD in SPSS, use the one-way between-subjects ANOVA utility (**Watch Video 7.10: [Tukey's \(-Kramer's\) HSD in SPSS](#)**). If you did the analysis correctly, you should get a table entitled 'Multiple Comparisons'. I have by-passed the interpretation of the ANOVA output, because Tukey-Kramer's HSD is a single-step procedure.

Multiple Comparisons

Dependent Variable: accuracy
Tukey HSD

(I) group	(J) group	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
federal agents	federal judges	11.00036*	3.60458	.013	1.6952	20.3055
	police officers	22.20000*	4.08863	.000	11.6453	32.7547
	clinical psychologists	10.90000*	3.36491	.007	2.2135	19.5865
federal judges	federal agents	-11.00036*	3.60458	.013	-20.3055	-1.6952
	police officers	11.19964*	3.05117	.002	3.3231	19.0762
	clinical psychologists	-.10036	1.97874	1.000	-5.2084	5.0077
police officers	federal agents	-22.20000*	4.08863	.000	-32.7547	-11.6453
	federal judges	-11.19964*	3.05117	.002	-19.0762	-3.3231
	clinical psychologists	-11.30000*	2.76392	.000	-18.4350	-4.1650
clinical psychologists	federal agents	-10.90000*	3.36491	.007	-19.5865	-2.2135
	federal judges	.10036	1.97874	1.000	-5.0077	5.2084
	police officers	11.30000*	2.76392	.000	4.1650	18.4350

*. The mean difference is significant at the 0.05 level.

SPSS reported all possible combinations of pairwise comparisons in the 'Multiple Comparisons' output, consequently, half of the information is redundant. For example, the first comparison, federal agents versus federal judges was associated with a mean difference of 11.00. In the next category below it, the comparison is switched around to federal judges versus federal agents (-11.00). There is no point in reporting all of these redundant results. Pick one of each only.

Quite bafflingly, SPSS does not report the q -values associated with the Tukey-Kramer's HSD statistics. Furthermore, the standard errors reported in the above table have nothing to do with Tukey-Kramer's HSD; they are the standard errors associated with Fisher's LSD! Consequently, it is not possible to calculate the q -values by dividing the mean differences by their corresponding standard error. To do so, it is necessary to calculate the Tukey-Kramer HSD standard errors "by hand". Fortunately, the p -values reported in the table above are, in fact, accurate Tukey-Kramer HSD based estimates. All of the mean comparisons were statistically significant, with the exception of the federal judges versus clinical psychologists comparison ($p = .999$). The p -value for that comparison was actually 1.00, but statisticians always round down to .999, because nothing is ever associated with 100% probability. Thus, the federal agents were statistically significantly better at detecting liars, in comparison to the federal judges, police officers, and clinical psychologists. The clinical psychologists and the judges were statistically significantly better at detecting lies, in comparison to the police officers.

SPSS also produced a second Tukey-Kramer's HSD table of results entitled 'Homogeneous Subsets'. There is nothing to be gained by interpreting this 'homogenous subsets' version of the Tukey's HSD. On top of not providing p -values, it is based on the version of Tukey's HSD that assumes equal sample sizes, which will be rarely the case in practice (and is certainly not the case in this example).

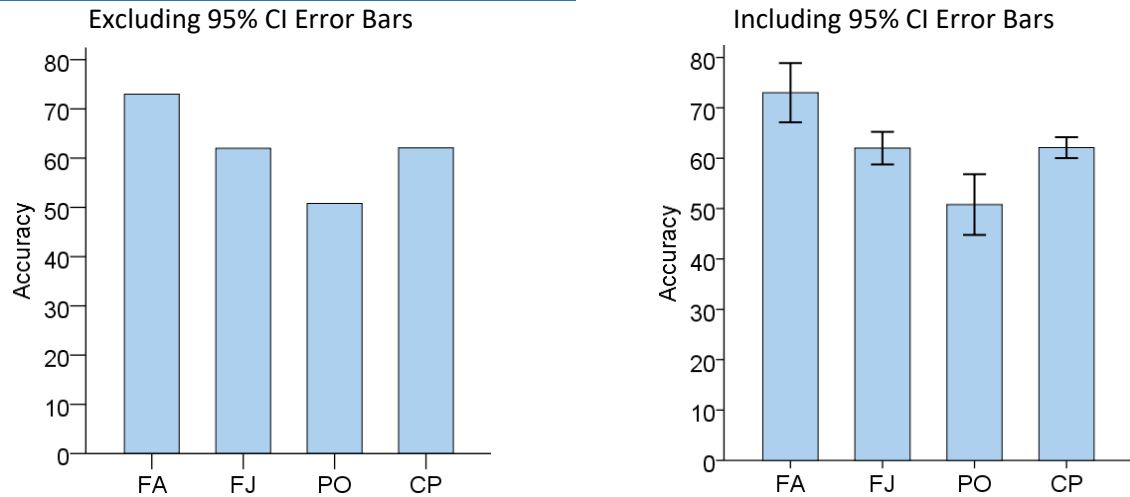
The Tukey-Kramer HSD is not a bad multiple comparison option, in most cases. However, it is known to be too conservative. By contrast, Fisher's LSD is too liberal for scenarios with more than three means. Fortunately, a statistician combined the virtues of the two tests into one. It is known as the Fisher-Hayter multiple comparison procedure. Unfortunately, it is not available in any software packages of which I am familiar. If you're not afraid of getting your hands a little dirty, I demonstrate how to conduct a Fisher-Hayter multiple comparison procedure in the Advanced Topics section of this chapter (with an Excel file to make things easy).

One-Way Between-Group ANOVA - Graphing Results: SPSS

There are several ways to create a bar-chart to represent the means associated with several groups. A key decision to be made is whether some sort of error bars should be included in the chart. Researchers variously include the standard deviations, the standard errors (equivalent to 68% confidence intervals), or 95% confidence intervals. In my opinion, if a researcher decides to use $\alpha = .05$ for the statistical analyses, it makes sense to add the 95% confidence intervals to the bar chart (not the standard deviations, or the standard errors). You should be mindful that 95% confidence intervals can intersect with each other, even when there is a statistically significant difference between the means. Usually, they can overlap by about 50% when $p = .05$ (Cummins, 2009).

Personally, I don't get much out of the inclusion of error bars of any sort to a bar chart. However, they are so commonly included that I have created two bar charts: one excluding the confidence intervals and one including the confidence intervals (See Figure C7.2).

Figure C7.2: Bar Charts Associated with the Catch A Liar Data



Note. FA = federal agent; FJ = Federal judge; PO = police officer; CP = clinical psychologist.

ANOVA: Variances Unequal

Many states/countries have laws against texting on a mobile phone while driving, presumably because texting increases the chances of getting into an accident. Alosco et al. (2012) wanted to test the hypothesis that texting and eating (not at the same time) would impact motor vehicle driving performance. To test the hypothesis, Alosco et al. (2012) had three groups of individuals drive a car in a simulator. During the driving exercise (about 20 minutes), a group of participants ($N = 45$) had to read and respond to 5 text messages on a mobile phone. In another group ($N = 45$), the participants had to eat two pre-packaged breakfast pastries and drink bottled water. Finally, a control group ($N = 96$) simply had to drive (no texting, no eating). Alosco et al. (2012) measured driving performance a number of different ways, one of which was the number of times the car veered off the side of the road. I generated data to replicate the Alosco et al. (2012) results very closely ([Data File: texting driving](#)).

The descriptive statistics associated with three groups are presented in the SPSS table entitled 'Statistics'. It can be seen that the texting group was associated with a mean of 3.49 road edge-excursions, which was numerically higher than the eating group ($M = 2.10$) and the control group ($M = 1.11$). The key question is whether these numerical differences were statistically significant (effect size estimation is important, too, of course). (**Watch Video 7.11: Separate Descriptive Statistics & Histograms in SPSS (Take 2)**)

Statistics			
Road-edge excursions			
texting	N	Valid	45
	Mean		3.49
	Std. Deviation		4.36
	Variance		18.98
	Skewness		2.65
	Kurtosis		9.80
eating	N	Valid	45.00
	Mean		2.10
	Std. Deviation		1.89
	Variance		3.57
	Skewness		1.63
	Kurtosis		2.46
control	N	Valid	96.00
	Mean		1.11
	Std. Deviation		1.94
	Variance		3.74
	Skewness		3.31
	Kurtosis		14.35

Before testing the hypothesis with a one-way between-subjects ANOVA, it is noteworthy that the variance associated with the texting group ($S^2 = 18.98$) was much larger than the variances associated with the eating ($S^2 = 3.57$) and control groups ($S^2 = 3.74$). As described above, the between-subjects ANOVA assumes homogeneity of variance, just like the independent samples t -test. Although the between-subjects ANOVA is robust to violations of the assumption of homogeneity of variance, this is only true when the sample sizes are equal. In this case, the control group ($N = 96$) is associated with a much larger sample size than the other two groups ($N = 45$). Therefore, one cannot rely upon the robustness of the regular between-subjects ANOVA, in this case.

Fortunately, the Welch's F -test and the Brown-Forsythe F -test are exceptionally good at keeping the alpha level at .05, when the variances are substantially different, and even when the sample sizes are unequal (Kohr & Games, 1974; Tomarken & Serlin, 1986). There is some evidence to suggest that the Welch's F -test will prove the best option, in most cases. Specifically, Tomarken and Serlin (1986) found that the Welch's F -test was associated with a Type I error rate close to .05 across various conditions of unequal variances and sample size differences (with 3 or 4 groups), whereas the Brown-Forsythe test yielded a Type I error rate closer to .06. Thus, as a general recommendation, the Welch's F -test should be preferred. Both F -tests can be performed as one-way between-subjects ANOVAs in SPSS.

Welch's *F*-test and Brown-Forsythe *F*-test: SPSS

In order to conduct a Welch's *F*-test or a Brown-Forsythe *F*-test in SPSS, use the one-way ANOVA utility (**Watch Video 7.12:** [Welch's and Brown-Forsythe's *F*-test in SPSS](#)). As can be seen in the SPSS table entitled 'Robust Tests of Equality of Means', the Welch *F*-test was statistically significant, $F(2, 82.33) = 8.75, p < .001$. The Brown-Forsythe *F*-test was also statistically significant, $F(2, 72.86) = 9.27, p < .001$. Thus, even though the assumption of homogeneity of variance was violated, and the sample sizes were unequal, the null hypothesis of equal means was rejected.

Robust Tests of Equality of Means				
Road-edge excursions				
	Statistic ^a	df1	df2	Sig.
Welch	8.745	2	82.325	.000
Brown-Forsythe	9.269	2	72.857	.000

a. Asymptotically *F* distributed.

It is possible to calculate a measure of effect size for the Brown-Forsythe *F*-test. If you're interested, I describe the procedure in the Advanced Topics section of this chapter.

Planned Comparisons

As mentioned above, multiple comparisons is a term I prefer to use in the context of testing the difference between several means across a series of statistical tests. Some multiple comparison procedures are single-step procedures (e.g., Tukey-Kramer's HSD; Games-Howell), while others are two-step procedures which require the observation of a statistically significant ANOVA (Fisher's LSD; Fisher-Hayter). When all possible mean comparisons are tested, the analysis is known as 'post-hoc' testing (a.k.a., 'unplanned comparisons'). For example, in the catch a liar study described above, all possible comparisons between the agents, judges, police officers, and clinical psychologists were tested with Tukey-Kramer's HSD (i.e., 6 tests). Such an analysis would be called 'post-hoc' testing by many researchers, even though I referred to it as a multiple comparison analysis.

In some cases, a researcher may have an interest in only a select number of mean comparisons, rather than all possible mean comparisons. If the researcher decides to conduct a select number of comparisons, these tests are known as 'planned comparisons'. A researcher cannot first look at the data and decide which planned comparisons are going to be performed. That would be cheating. Instead, the researcher should decide on the planned comparisons prior to the collection of the data. The main benefit of testing hypotheses via the planned comparison approach is that it is more powerful: that is, the chances of correctly rejecting a null hypothesis increases. A planned comparison approach is more powerful, because fewer statistical analyses are performed on the data.

Keppel (1991) suggested that a total of $k-1$ planned comparisons can be performed, without any requirement to employ any adjustments to control for an increase in familywise

error rate. They symbol k represents the number of groups in the analysis. In the catch a liar study, there were four groups. Thus, according to Keppel (1991), $4 - 1 = 3$ planned comparisons could be performed, without the obligation of employing a procedure to keep the familywise error rate from increasing beyond per contrast alpha (i.e., .05). I discuss planned comparisons in more detail in the Advanced Topics section of this chapter.

Pairwise and Non-Pairwise Comparisons

Thus far, I have focused exclusively upon what are known as pairwise comparisons. Pairwise comparisons involve the comparison of two means only. However, there is another type of comparison known as a non-pairwise comparison. A non-pairwise comparison includes information from three or more means, simultaneously. In a typical non-pairwise comparison, two means (combined) are compared against another mean. For example, a researcher might compare a control group against a combined low-dose and high-dose treatment group. The overwhelming vast majority of mean comparisons reported in the literature are consistent with a pairwise comparison approach. Multiple comparison procedures such as Tukey-Kramer's HSD, Fisher-Hayter's q -test, and Fisher's LSD were designed to control the familywise error rate for a collection of pairwise comparisons only. If a combination of pairwise and non-pairwise comparisons are desired to be performed, alternative multiple comparison procedures need to be considered.

A commonly observed recommendation suggests that the Bonferroni correction should be applied when testing a collection of planned pairwise and non-pairwise comparisons. However, if a true fishing expedition is undertaken with a collection of unplanned pairwise and unplanned non-pairwise comparisons, then the Scheffé multiple comparison procedure should be considered for the purposes of keeping the familywise error rate near .05. Most statistical software packages include the Scheffé multiple comparison procedure, including SPSS. As a single step-procedure, Scheffé is similar to the Tukey-Kramer procedure. However, Scheffé's procedure is based on the F -distribution, rather than the q -distribution.

Dunnett's Multiple Comparison Procedure

Dunnett's multiple comparison procedure is an attractive single-step procedure. Dunnett's multiple comparison procedure was designed specifically for the case where only pairwise comparisons with a control group (or a comparison group) are of interest. For example, a researcher may have three groups: control group, drug A group, and drug B group. If the only comparisons of interest are drug A versus control, and drug B versus control, the most powerful statistical procedure to employ, while also keeping the familywise error rate at .05, is Dunnett's multiple comparison procedure. Keep this multiple comparison in mind, when you have such a research scenario. It can be performed within most software packages, including SPSS (**Watch Video 7.13:** [Dunnett's Multiple Comparison Test in SPSS](#)).

Advanced Topics

Calculate Eta-Squared from F -value

There are many occasions where I read a paper that includes the results of an ANOVA; however, the paper does not include a measure of effect size for the ANOVA. Fortunately, η^2 can be calculated from an F -value and the degrees of freedom, as depicted in the following formula:

$$\eta^2 = \frac{F * df_{\text{Between}}}{F * df_{\text{Between}} + df_{\text{Within}}} \quad (11)$$

For the alcohol and memory study example described in the main section of this chapter, the F -value was 15.25 and the degrees of freedom were 2 and 69. Thus, η^2 was estimated at .307, which, you may recall, is exactly the same value obtained from the typically observed SS_B/SS_T

$$\eta^2 = \frac{15.25 * 2}{15.25 * 2 + 69}$$

$$\eta^2 = \frac{30.50}{99.50} = .307$$

The fact that η^2 can be calculated from the obtained F -value and the degrees of freedom (df) has implications for estimating η^2 for a slightly different one-way between-subjects ANOVA described below. Specifically, the Browne-Forsythe F -test, which does not assume homogeneity of variance or equal sample sizes.

Calculate Confidence Intervals for Eta-Squared & Omega-Squared (Forthcoming)

Calculate Tukey-Kramer q -values

As mentioned in the Foundations section of this chapter, SPSS does not report the q -values associated with the Tukey-Kramer HSD mean comparisons. However, they can be calculated “by hand” with the following formula:

$$\text{Tukey - Kramer HSD } q = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{MS_{\text{Within}}}{2} \left(\frac{1}{N_1} + \frac{1}{N_2} \right)}} \quad (12)$$

where N_1 and N_2 are equal to the sample size of each group (they are not assumed to be equal across all groups). The calculated q -values are tested for statistical significance against the q distribution with k (number of groups) and df_{Within} degrees of freedom. You can use the following Excel function to calculate Tukey-Kramer HSD q -values:

$$=(A2-B2)/SQRT((E2/2)*((1/C2)+(1/D2)))$$

The main reason I would report the q -value in a thesis or article is thoroughness. That is, if you relied upon the SPSS output, you would only be able to report the p -value. I can't imagine someone doing that justifiably for a t -test, so why do it for a q -test? The same Excel function is described in more detail in the section below devoted to the Fisher-Hayter q -statistic.

Fisher-Hayter Multiple Comparison Procedure

Hayter (1986) proved that the Tukey-Kramer HSD multiple comparison was conservative. Consequently, Hayter (1986) developed a modification of the Tukey-Kramer HSD that was less conservative, but still kept the familywise error rate close to .05. The formula used to calculate the Fisher-Hayter q -value is identical to the Tukey-Kramer q formula:

$$\text{Fisher - Hayter } q = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{MS_{\text{Within}}}{2} \left(\frac{1}{N_1} + \frac{1}{N_2} \right)}} \quad (13)$$

The calculated q values are tested against the relevant critical q -distribution value based on $k-1$ and df_{Within} . The first key difference between the Fisher-Hayter q -test procedure and the Tukey-Kramer HSD q -test is that the Fisher-Hayter test requires the first-step of a statistically significant ANOVA. Consequently, the Fisher-Hayter q -test is a two-step procedure, just like Fisher's protected LSD multiple comparison procedure. However, in contrast to Fisher's LSD, the Fisher-Hayter multiple comparison will keep the familywise error rate closer to .05 when applied to analyses which include four or more groups.

The second key difference is that the value of a $k-1$ is used, instead of k (k = number of groups), in order to identify the critical q -value for statistical significance. The reduction of k by 1 increases the power of the Fisher-Hayter test, in comparison to the Tukey-Kramer HSD. Unfortunately, SPSS does not offer a menu option to perform the Fisher-Hayter multiple comparison procedure. Fortunately, however, with a bit of extra work, it is possible to get SPSS to calculate the p -values associated with the Fisher-Hayter q -values. As the Fisher-Hayter q -test will be typically between 2% and 9% more powerful than Tukey's HSD (Seaman et al., 1991), it is worth the extra effort, in my opinion.

Fisher-Hayter Multiple Comparison Procedure: SPSS

First, a between-subjects ANOVA must be performed. If the F -value is statistically significant, it is appropriate to proceed with the Fisher-Hayter q test calculations. However, if the F -value is not statistically significant, the analysis should be terminated. If the ANOVA is statistically significant, one can calculate the q -values associated with each pairwise comparison. In this 'catch a liar' study example with four means (federal agent, federal judge, police officer clinical psychologist; [Data File: catch a liar](#)), there are six pairwise comparisons. For each comparison, the standard error of the difference between the means needs to be calculated

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with the Fisher-Hayter q -test formula presented above. The q -value can be calculated in excel with the following function:

$$=(A2-B2)/SQRT((E2/2)*((1/C2)+(1/D2)))$$

where the following values apply: A2 = Mean 1; B2 = Mean 2; C2 = Sample Size group 1; D2 = Sample Size Group 2; E2 = Mean Square Within (**Watch Video 7.14: [Fisher-Hayter Multiple Comparison Test \(SPSS & Excel\)](#)**).

For the federal agents versus federal judges comparison, I obtained a Fisher-Hayter q -value of 4.315741, based on the application of the above excel function:

	A	B	C	D	E	F	G	H	I	J
1	Mean1	Mean2	N1	N2	MSWithin		M Difference =	11.00	q=	4.315741
2	73.00	62.00	23	84	234.6		Standard Error =	2.548809		

In order to calculate the p -value associated with the q -value of 4.3157, I used the following function in SPSS within the Compute utility (**Watch Video 7.15: [How to obtain a p-value for a q-test value in SPSS](#)**):

$$(1-\text{CDF.SRANGE}(q, 3, 348))$$

The q -value corresponds to the obtained q -value reported by the Excel function and inputted into the first column of SPSS (label the column 'p'). The value of 4 corresponds to the number of groups minus one ($k - 1$) associated with the ANOVA. The value of 348 corresponds to the within-groups degrees of freedom. After I ran the above compute statement, I obtained a p -value of .0069.

	q	p
1	4.3157	.006905770

Thus, with respect to the federal agents versus federal judges mean comparison, the q -value of 4.3157 was found to be statistically significant, $p = .007$. For the purposes of comparison, I'll note that SPSS estimated the Tukey-Kramer HSD p -value for the same mean comparison at .013. Thus, as expected, the Fisher-Hayter q -test approach was more powerful. For the purposes of comparison, I have reported all of the Tukey-Kramer HSD and Fisher-Hayter HSD p -values for all of the multiple comparisons in the catch a liar study (see Table C7.7). Recall that the Tukey-Kramer HSD and the Fisher-Hayter test share exactly the same q -value. However, it can be seen that the Fisher-Hayter q -test procedure was consistently more powerful (i.e., smaller p -values).

Table C7.7. Multiple Comparison Results

			q	T-K HSD	F-H
Federal Agent	vs.	Federal Judge	4.316	.013	.007
Federal Agent	vs.	Police Officers	7.679	.000	.000
Federal Agent	vs.	Psychologist	4.581	.007	.004
Federal Judge	vs.	Police Officer	5.191	.002	.001
Federal Judge	vs.	Psychologist	0.643	1.00	.892
Police Officer	vs.	Psychologist	-5.782	.000	.000

Note. T-K HSD = Tukey-Kramer HSD; F-H = Fisher-Hayter; the Tukey-Kramer HSD and the Fisher-Hayter q statistics are based on the same formula, consequently, only one row of q statistics has been reported in the table above.

Appropriate Confidence Intervals for Multiple Comparisons

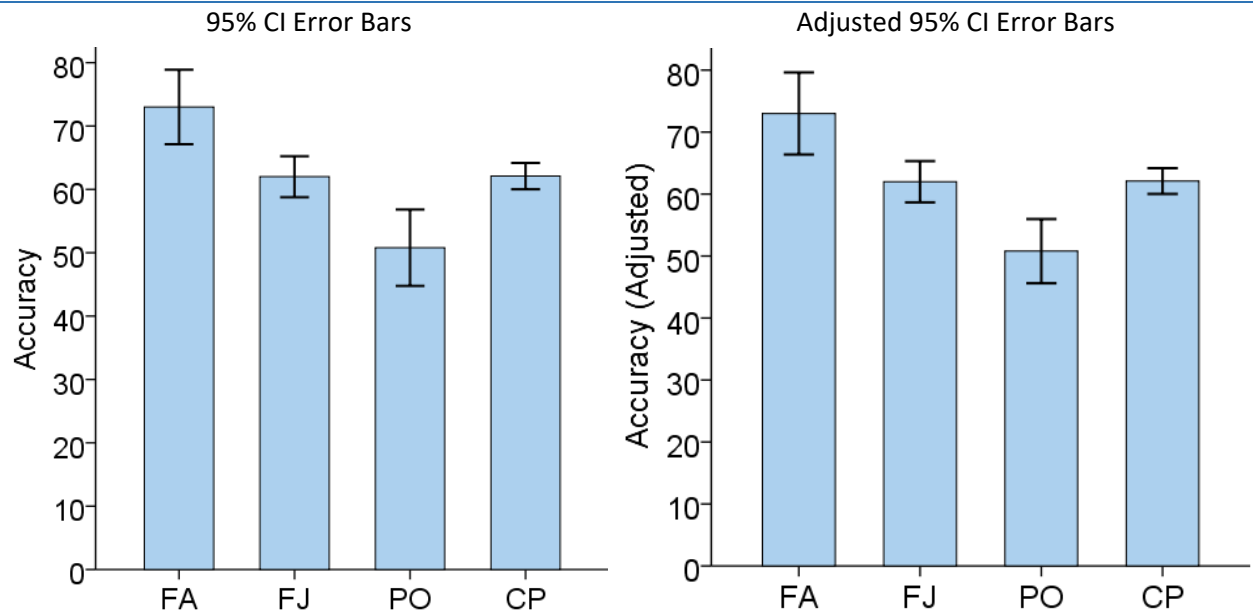
Technically, most software packages display incorrect standard errors and/or confidence intervals for bar charts that compliment an ANOVA or a multiple comparison procedure. The reason is that ANOVA and most multiple comparison procedures use a **pooled error term**. However, the confidence intervals/error bars displayed in Figure C7.2 of this chapter were estimated separately for each of the four groups. Consequently, it would appear to be the case that most researchers include the inappropriate error bars (confidence intervals) in their bar charts.

If you are really anal about these sorts of things (not a bad thing), you can create the appropriate 95% CI error bars by calculating the average of the standard deviations ($\overline{SD}$) associated with the groups included in the analysis (i.e., pooled standard deviation). The quickest way to do that is to use the following formula:

$$\overline{SD} = \sqrt{MS_W} \quad (14)$$

...where MS_W is from the ANOVA summary table. Then, transform the raw scores into z-scores. Finally, convert the z-scores to have means which correspond to the raw score means and standard deviations which correspond to the average standard deviation. Use the converted scores as data to create the bar charts with 95% CI error bars. The procedure is not as complicated as it sounds. (**Watch Video 7.16: [Appropriate Error Bars & CIs for Oneway Between-Subjects ANOVA](#)**). With respect to the 'catch a liar' study, the pooled SD was calculated at $\sqrt{234.603} = 15.317$. The adjusted bar chart can be seen in Figure C7.3 (right-side). Admittedly, there's not much difference between the two charts. Perhaps most notably, the FA group adjusted 95% confidence interval is a little larger and the PO group adjusted 95% confidence interval is a little smaller.

Figure C7.3: Unadjusted and Adjusted Bar Charts Associated with the Catch A Liar Data



Note. FA = federal agent; FJ = Federal judge; PO = police officer; CP = clinical psychologist.

Welch's F -test and Brown-Forsythe (B-F) F -test: Effect size

Although SPSS does not report η^2 with the Welch's and Brown-Forsythe (B-F) F -tests, or even the sums of squares, an estimate of η^2 can be approximated based on the Brown-Forsythe F -value and the degrees of freedom associated with the regular one-way between-subjects ANOVA (not the adjusted degrees of freedom). The B-F η^2 formula is below:

$$\text{B-F } \eta^2 = \frac{F * df_{\text{Between}}}{F * df_{\text{Between}} + df_{\text{Within}}} \quad (15)$$

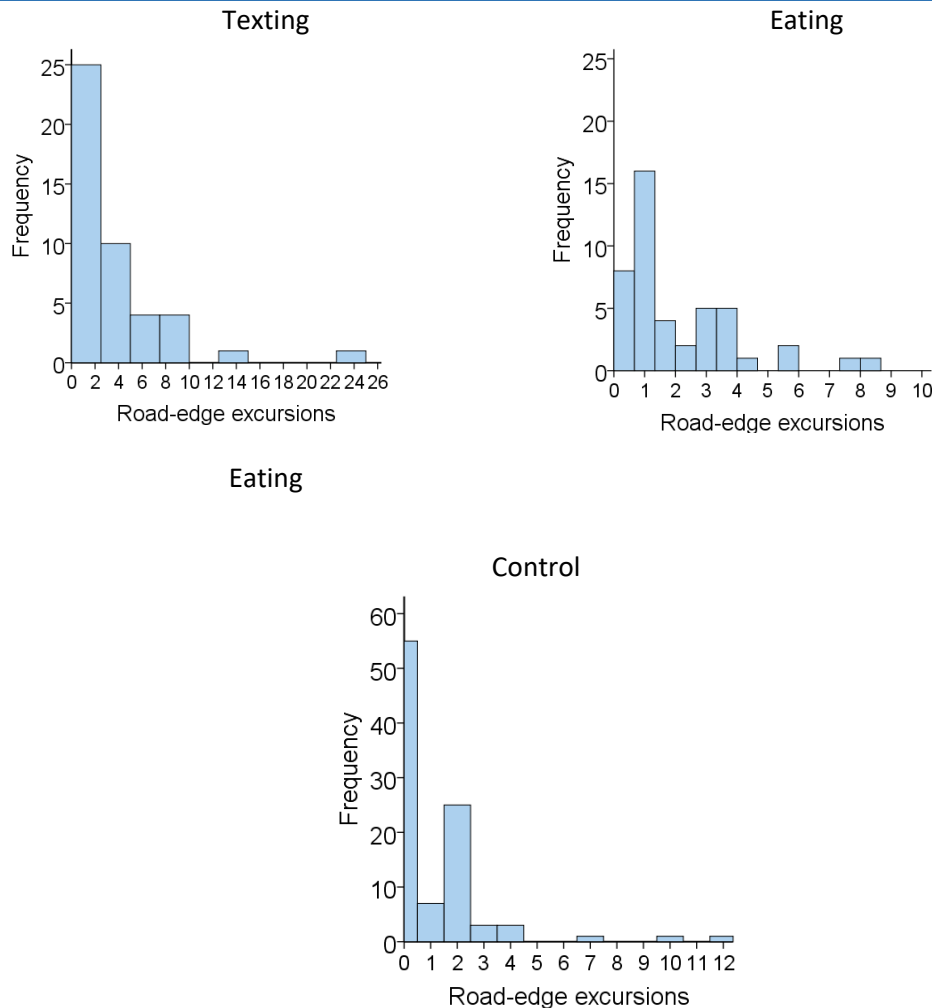
Recall in the 'road excursions' study that the null hypothesis was rejected, $F(2, 72.857) = 9.269$, $p < .001$. Furthermore, the degrees of freedom for the 'regular' ANOVA were 2 and 183. Solving the above formula with the relevant values yielded an η^2 estimate of .092.

$$\text{B-F } \eta^2 = \frac{9.27 * 2}{9.27 * 2 + 183} = \frac{18.54}{201.54} = .092$$

Thus, 9.2% of the variance in road-edge excursions was accounted for by the three driving groups. Thus, the above use of the Brown-Forsythe F -value to estimate eta-squared has never been proven, to my knowledge. It simply makes sense to me. Hopefully, someone will investigate this issue, one day.

Although Alosco et al (2012) did not report any information about the distributions of the scores they collected, it is almost certainly the case the data were substantially positively skewed. Consider that the standard deviations were all relatively large in comparison to the means, and that the lowest possible score was zero. Based on my simulation of the data, I obtained the following distributions (Figure C7.4) and descriptive statistics.

Figure C7.4. Histograms



Clearly, the data were substantially non-normally distributed. Furthermore, the levels of skew and kurtosis exceeded the rule of skew $< |2.0|$ and kurtosis $< |9.0|$ for parametric statistics. What's more, it looks like there were one or more outliers in the data. Many statisticians would be tempted to move onto a non-parametric statistic in this case, that is, a statistic that would convert the data into ranks. However, I would much prefer to stick with the raw data, because it allows for more interesting inferences. A bootstrapped version of the Welch

or Brown-Forsythe adjusted F -test would be really useful, in this case. Unfortunately, SPSS does not offer the opportunity to conduct a bootstrapped version of these two tests.

In this situation, that is, comparing more than two between-subjects means with unequal variances, unequal sample sizes, and very non-normally distributed data, I would perform a Welch's F -test or a Brown-Forsythe adjusted F -test with alpha specified at .01. That is, a more conservative alpha level than .05. Based on my familiarity with the simulation research (e.g., Harwell, Rubenstein, Hayes, & Olds, 1992; Zimmerman & Zumbo, 1993), an alpha level of .01 should keep the chances of committing a Type I error at less than .05, in the vast majority of cases. In the Alosco et al (2011) 'road excursions' example study, the Brown-Forsythe p -value was less than .001. Consequently, as .001 is less than .01, I would reject the null hypothesis of equal means.

Arguably, a more attractive alternative to the above would be to use a single-step multiple comparison procedure that does not assume homogeneity of variance or equal sample sizes. The procedure is known as the Games-Howell t -test. When the Games-Howell procedure is estimated via bootstrapping, it would also not assume normally distributed data (see next Advanced Topic).

Games-Howell Multiple Comparison Procedure

The statistically significant Welch F -test and Brown-Forsythe F -test ANOVAs simply tells us that at least one comparison amongst the means is statistically significant. However, most researchers would want to know which means specifically are statistically significantly different from each other. As the data were associated with unequal variances, a solid option is to test the multiple comparisons with the Games-Howell multiple comparison test, as it does not assume equal variances, nor does it even assume equal sample sizes. Even for very seriously violated scenarios (largest to smallest variance ratio = 10; largest to smallest sample size ratio = 1.85), the Games-Howell procedure will keep the familywise alpha level close to .04 (Dunnett, 1980). So, if anything, the Games-Howell test is a little conservative. Finally, the Games-Howell procedure is a single-step procedure. Consequently, one does not need to observe a statistically significant ANOVA (or Welch's F -test or Brown-Forsythe F -test) first.

As the data are very non-normally distributed in this example, I would estimate the Games-Howell t -values and p -values via bootstrapping, which does not assume normally distributed data (or even the absence of outliers).

Games-Howell Multiple Comparison Procedure: SPSS

To conduct a Games-Howell multiple comparison analysis in SPSS, select the Games-Howell post-hoc option in the one-way between-subject ANOVA utility. Also, ensure the

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bootstrapping utility is activated (1000 samples)⁷, if you have data that are substantially non-normally distributed (skew > |2.0|; kurtosis > |9.0|). When I ran the analysis on the current example data, an SPSS table entitled ‘Bootstrap for Multiple Comparisons’ was produced (**Watch Video 7.17:** [Ultra Robust Multiple Comparisons: Games Howell w/ Bootstrap](#)):

Bootstrap for Multiple Comparisons						
Dependent Variable: Road-edge excursions						
Games-Howell						
(I) group	(J) group	Mean Difference (I-J)	Bootstrap ^a			
			Bias	Std. Error	BCa 95% Confidence Interval	
					Lower	Upper
texting	eating	1.39444	-.05775	.70449	.20004	2.66094
	control	2.37986	-.05370	.68449	1.27658	3.56584
eating	texting	-1.39444	.05775	.70449	-3.09274	.04326
	control	.98542	.00405	.33971	.32571	1.66985
control	texting	-2.37986	.05370	.68449	-4.18972	-1.05246
	eating	-.98542	-.00405	.33971	-1.65170	-.34068

a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

Unfortunately, SPSS does not report the t -value (or the p -value) associated with the Games-Howell statistics.⁸ You can, however, estimate the t -value yourself by dividing the mean difference by the standard error of the difference between the means. I have reported the results in Table C7.8.

Table C7.8. Statistics Associated with Games-Howell Comparisons

			ΔM	SE	t -value	Cohen's d	p
Texting	vs.	Eating	1.394	.705	1.97	.72	<.05
Texting	vs.	Control	2.380	.685	3.47	1.23	<.05
Eating	vs.	Control	.985	.339	2.91	.51	<.05

Note. Cohen's d values were calculate by using only the control group SD.

As all of the t -values were greater than |1.96|, all three mean comparisons were statistically significant ($p < .05$). Furthermore, the 95% confidence intervals associated with the difference between the means did not intersect with zero, which supports further that the mean

⁷ If you have a large number of groups to compare, say 5 or more, the bootstrapping analysis can take a fair bit of time (even just three groups takes a couple of minutes). Reducing the number of bootstrapped samples to 500 would be an acceptable decision in such a case, which should reduce the running time of the analysis. I wouldn't use less than 500 re-samples, however.

⁸ Many sources describe the Games-Howell procedure as a q -statistic, however, the formula used by SPSS renders the Games-Howell procedure such that it follows the t -distribution. Consequently, it is more accurate to refer to the SPSS Games-Howell results as t -tests.

differences were statistically significant. There was one exception: although the 95% confidence interval difference between the texting and eating means did not intersect with zero (.200/2.66), the difference between eating and texting did intersect with zero very slightly (-3.09/.04). The reason this has happened is that SPSS used a different set of bootstrapped samples to estimate the 'bias-corrected accelerated' (BCa) confidence intervals. That is, for each analysis, the bootstrapped samples are selected at random. Consequently, the samples for the texting versus eating comparison and the eating versus texting comparison were different. I'll also note that the bootstrap algorithm used by SPSS to estimate the standard error of the difference between the means is different to the algorithm used to estimate the BCa 95% confidence intervals. Consequently, the results may not be consistent completely. I would probably trust the 'bias-corrected accelerated' results over the other results in most cases. However, they will rarely disagree with each other, in practice, with respect to whether you should reject the null hypothesis or not. Ultimately, the texting versus eating comparison is right on the line, irrespective of the bootstrapping approach taken. Note that you are unlikely to get the same exact results as those reported above, because your 1000 re-samples will not be the same as mine. The 1000 re-samples are selected randomly by the program.

Because the variances were unequal, Cohen's d was calculated by using the control group standard deviation (1.94) in the denominator of the Cohen's d formula. Technically, this version of Cohen's d is known as Glass's Δ . It can be seen that the texting versus control group comparison was associated with a large effect size, according to Cohen's (1988) guidelines. In plain language, on average, texting increased the number of occasions a person veered off the road by about approximately a factor of three ($3.49 / 1.11 = 3.14$). Alosco et al (2012) also reported that texting while driving increased the chances of hitting a pedestrian. Clearly, texting while driving is a dangerous activity.

Planned Comparisons (Contrasts)

As mentioned in the Foundations section of this chapter, it is easy to get confused by reading the multiple comparisons literature. I believe part of the confusion is due to the large number of terms that are used sometimes interchangeably and/or sometimes ambiguously. For the purposes of clarity, I consider the words 'comparison' and 'contrast' as entirely synonymous (they mean the same thing). Specifically, I consider all tests of the difference between means a comparison. I also consider them a contrast. However, in the literature, you will see people reserve use of the word contrast to describe non-pairwise comparisons only (i.e., not pairwise comparisons). In my view, both pairwise and non-pairwise comparisons are contrasts. Allow me to demonstrate.

When people use the word contrast, they refer to tests of the difference between means which are associated with 'weights'. However, even the simplest test of the difference between two means has 'weights', whether they are explicitly specified or not. Consider, for

example, the typical independent samples t -test. In the Foundations section of the test of the difference between two means chapter, the independent samples t -test was formulated as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{N_1} + \frac{S_2^2}{N_2}}} \quad (16)$$

Although it may not be apparent, the above formula has implied weights. To be placed more clearly within the context of a contrast, the numerator portion of the above well-loved formula can be re-specified as:

$$\psi = [(+1)(\bar{X}_1)] + [(-1)(\bar{X}_2)] \quad (17)$$

Although the above formula looks much more complicated than the numerator portion of the conventional t -test formula, it's really exactly the same. The $[(+1)(\bar{X}_1)]$ term is really just mean 1, right? Specifically, mean 1 multiplied by +1 gives mean 1. Simple. A key difference, however, is that the contrast adds the means together, rather than necessarily subtracts them. That is, you should notice that there is a + sign that separates the two terms, rather than a – sign. The second term in the contrast is $[(-1)(\bar{X}_2)]$. Thus, mean 2 is multiplied by a negative value. Specifically, -1. Thus, within the contrast, mean 2 turns into a negative value. Because there is a plus sign in-between the two terms, and the second mean has been weighted with -1, the actual procedure involves subtracting mean 2 from mean 1, just as specified in the typical independent samples t -test. As mentioned, people would refer to the expression in equation 16 as a contrast, because it has the weightings (+1 and -1). However, the typical independent samples t -test has implied weightings; they're just not explicitly specified.

By this point, you might think that this whole contrast business was designed to complicate the lives of students; but that is not quite true. An understanding of the specification of contrasts opens up the possibility of testing a whole new world of comparisons: non-pairwise comparisons. There is a huge number of different types of non-pairwise comparisons that can be specified, especially when the number of means included in the analysis is more than three. However, I would say that researchers would be well served by knowing only three different types of non-pairwise comparisons/contrasts: (1) linear contrasts (positive and negative); (2) quadratic contrasts (upward and downward), and (3) combined treatments versus control group contrasts. I cover all three types of contrast in the next few sections of the Advanced Topics.

For your convenience, I have provided in Table C7.9 the contrast weightings that can be used to test many of the commonly observed contrasts in the behavioral sciences. The number of groups ranges from only 3 to 5. I suspect this will suffice for the vast majority of cases. You may detect a pattern in the weightings: the weightings for any given contrast always sum to zero. That is a rule for the specification of an appropriate contrast (**Watch Video 7.18: [What are contrast weightings in planned comparisons?](#)**).

Table C7.9. Commonly Used Contrast Weightings

Number of Groups	Type of Contrast	Weightings				
3	Linear	-1	0	1		
	Quadratic	-1	2	-1		
4	Linear	-3	-1	1	3	
	Quadratic	-1	1	1	-1	
	Cubic	-1	3	-3	1	
5	Linear	-2	-1	0	1	2
	Quadratic	2	-1	-2	-1	2
	Cubic	-1	2	0	-2	1
	Quartic	1	-4	6	-4	1

Linear Contrasts

At the beginning of this chapter, I presented a one-way between-subjects ANOVA to test the difference between three memory span mean across three groups: low dose alcohol, medium dose alcohol, and high dose alcohol. The analyses were based on the results reported by Peterson et al. (1990). Instead of applying a conventional approach to testing the null hypothesis with these data, one could employ a contrast analysis approach. A contrast approach would be more powerful statistically. Theoretically, one might hypothesize that the trend of the memory span means would be linear: that is, systematically and consistently upward or downward. In the case of Peterson et al. (1990), we would expect higher doses of alcohol to impact memory span ability negatively ([Data File: alcohol_memory](#)). Thus, we would expect a downward trend in the memory span means. Thus, based on the values reported in Table C7.9, we could use the following contrast weightings: +1, 0, -1. When I applied those contrast coefficients in SPSS, I obtained the following results in a table entitled 'Contrast Tests' (**Watch Video 7.19:** [Linear Contrast Analysis in SPSS](#)):

Contrast Tests						
Contrast			Value of Contrast	Std. Error	t	Sig. (2-tailed)
memory	Assume equal variances	1	-8.7100	1.66247	-5.239	.000
	Does not assume equal variances	1	-8.7100	1.52666	-5.705	.000

Based on a t -value of -5.239 and 69 degrees of freedom, the null hypothesis of no downward linear trend was rejected, $t(69) = -5.24, p < .001$. The negative t -value indicates that the means decrease in magnitude. Based on the contrast coefficients, had the t -value been reported as positive, it would have suggested that the means increased in magnitude, which would have been inconsistent with the hypothesis.

Contrasts and Effect Size

As discussed by Furr (2004), there are three approaches to representing effect size in the context of contrast analyses. The three approaches are essentially: (1, $r^2_{\text{effect size}}$) the squared correlation between the contrast weights and the individual case scores; (2, r^2_{alerting}) the squared correlation between the contrast weights and the group means; and (3, r^2_{contrast}) the unique squared correlation between the contrast weights and the individual cases scores, controlling for the variance accounted for by other contrast weightings (e.g., linear, quadratic, cubic, etc.). In practice, researchers rarely use the third approach to estimating the effect size associated with a contrast analysis, although I can see its potential use, in some instances. In my view, the first option should probably be considered the most appropriate measure of effect size for a contrast analysis, as it is most closely aligned with the hypothesis that is actually tested by the t -value (or F -value). Furthermore, the second option tends to inflate how big of an effect there may be in the data, as it ignores the total variance within the data.

In order to calculate $r^2_{\text{effect size}}$ the most accurately, you would need to create another variable (column of data) to represent each case's contrast weighting (based upon which group they belong to). Then, calculate a Pearson correlation between the contrast weighting variable and the dependent variable. The squared correlation would represent the percentage of variance accounted for in the dependent variable by the contrast. When I did so, I obtained a Pearson correlation of .525, which corresponds to $r^2_{\text{effect size}}$ of .276.

A more efficient method is to use the following formula:

$$r^2_{\text{effect size}} = \frac{t^2}{F_{\text{Between}} * df_{\text{Between}} + df_{\text{Within}}} \quad (18)$$

where t^2 is derived from the contrast analysis and F_{Between} , df_{Between} , and df_{Within} are derived from a regular oneway between-subjects ANOVA (with 'memory' as the dependent variable and 'dosage' as the independent variable). Thus, for the alcohol dosage and memory study example, the $r^2_{\text{effect size}}$ was estimated at .275, which is essentially identical to the squared Pearson correlation I obtained above:

$$r^2_{\text{effect size}} = \frac{5.239^2}{15.250 * 2 + 69} = .275$$

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In contrast to $r^2_{\text{effect size}}$, I estimated r^2_{alerting} at .901 (based on Pearson $r = .949$ between $X = +1, 0, -1$ and $Y = 27.71, 19.93, \text{ and } 13.06$), which implies that 90.1% of the variance in the three memory means was accounted for by the three contrast weightings. Again, in my view, this overstates the magnitude of the effect. However, I do acknowledge that, at the group level, ignoring individual differences, r^2_{alerting} does represent an effect associated with the contrast analysis.

Quadratic Contrasts

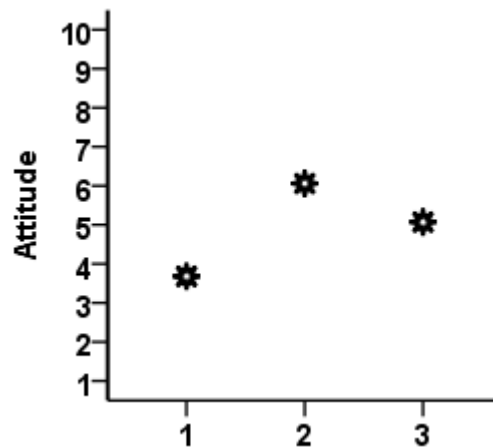
If ads have to be placed on a webpage, what sort do people like to see most? Moore, Stammerjohan, and Coulter (2005) were interested in this question. They speculated that ads that have nothing to do with the content of the webpage (incongruent) would be viewed relatively unfavorably (“waste of time”). Additionally, ads that were related very closely to the content of the webpage would also be viewed relatively unfavorably (perhaps a little “try hard”). The sweet spot, one may hypothesize, are ads that are moderately congruent with the content of the webpage. To test this hypothesis, Moore et al. (2005) collected data from 90 participants who rated the degree to which they were favorable to an ad (on a 9-point Likert scale; higher scores indicated higher favorability). One third of the participants ($N = 30$) viewed a webpage with an incongruent ad, one third ($N = 30$) viewed a webpage with a moderately congruent ad, and one third ($N = 30$) viewed a webpage with a congruent ad.

Moore et al. (2005) obtained the following three means from the three groups: incongruent $M = 3.68$, moderately congruent $M = 6.05$, and congruent $M = 5.07$. Clearly, there was not a linear increase in the means. Instead, the pattern of the means appears to be consistent with a quadratic effect. I have simulated some data to correspond to the results reported by Moore et al. (2005) ([Data File: ad congruency websites](#)). Furthermore, I tested statistically a hypothesis consistent with a quadratic effect, such that the webpage with a moderately congruent ad yielded, on average, the most favorable ratings (i.e., inverted U-shape). Recall that a quadratic contrast is consistent with one “bend” in the pattern of the means. When I ran the analysis on the current example data, an SPSS table entitled ‘Anova’ was produced (**Watch Video 7.20: [Non-Linear \(Quadratic\) Contrast Analysis in SPSS](#)**):

ANOVA							
attitude							
			Sum of Squares	df	Mean Square	F	Sig.
Between Groups	(Combined)		85.766	2	42.883	24.047	.000
	Linear Term	Contrast	28.982	1	28.982	16.252	.000
		Deviation	56.785	1	56.785	31.843	.000
	Quadratic Term	Contrast	56.785	1	56.785	31.843	.000
Within Groups			155.144	87	1.783		
Total			240.910	89			

It can be seen that the 'Quadratic Term Contrast' was associated with a statistically significant effect, $F(1, 87) = 31.84, < .001, \eta^2 = .236$. Thus, the null hypothesis of a purely linear effect can be rejected. As can be seen in Figure C7.5, there was one bend in the pattern of means which was consistent with an inverted-U shape. Thus, the hypothesis that the relationships between ad content and web page content would peak at moderate congruency was supported. Naturally, it would be tempting to conduct the multiple comparisons between the relevant means to determine whether the moderately congruent condition yielded statistically significantly more favorable ratings than the other two conditions. However, doing so would essentially defeat the purpose of conducting a dedicated, quadratic contrast analysis: to test a theoretically driven hypothesis with an efficient statistical analysis.

Figure C7.5. Plot of Means Attitudes Toward Ad Congruency



Notes. 1 = Incongruent; 2 = Moderately incongruent; 3 = Congruent.

Combined Treatments versus Control Group Contrasts

In the Foundations section of this chapter, I described the distinction between a pairwise comparison and a non-pairwise comparison. I encourage you to consult the relevant section of this chapter to refresh your memory, if required. Here, I provide a demonstration of perhaps the most typical non-pairwise contrast.

Suppose a pharmaceutical company conducted a clinical trial study to evaluate the efficacy of two different drugs designed to lower LDL cholesterol (the “bad” cholesterol). In one group, the participants took drug A ($N = 10$); in another group, the participants took drug B ($N = 10$); finally, a control group did not take either of the drugs ($N = 10$). After six months, all 30 participants had their LDL cholesterol levels measured. I have simulated some data to provide an interesting example to exemplify the advantage of having knowledge about non-pairwise contrasts ([Data File: cholesterol drugs](#)). First, I tested the typical equality of means (pairwise) null hypothesis with a regular one-between-subjects ANOVA. As can be seen in the SPSS table entitled ‘ANOVA’, the null hypothesis was not rejected, $F(2, 29) = 2.58, p = .095, \eta^2 = .160$.

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ANOVA

LDL

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	3845.000	2	1922.500	2.578	.095
Within Groups	20138.062	27	745.854		
Total	23983.062	29			

However, as can be seen in the Descriptive statistics table, the pattern of means suggest that there may have been some sort of effect of drugs versus placebo. That is, both Drug A and Drug B were associated with numerically lower LDL means than the control group. However, the non-significant ANOVA F -value of 2.578 implies that none of the individual mean comparisons are statistically significantly different from each other. There remains the possibility, however, that a non-pairwise comparison may be statistically significant.

Descriptives

LDL

	N	Mean	Std. Deviation	Std. Error
Drug A	10	106.00	27.90	8.82
Drug B	10	105.00	27.20	8.60
Control	10	129.50	26.82	8.48
Total	30	113.50	28.76	5.25

In this context, perhaps the most attractive non-pairwise comparison would be to test a combined Drug.A and Drug.B mean against the control group mean. The contrast would correspond to expression (17). Notice how the Drug.A and Drug.B means have been given “half” weightings ($-.50$) and the control group has been given a full weighting. Notice also that the contrast weighting sum to zero, which must be observed in order for a contrast to be valid.

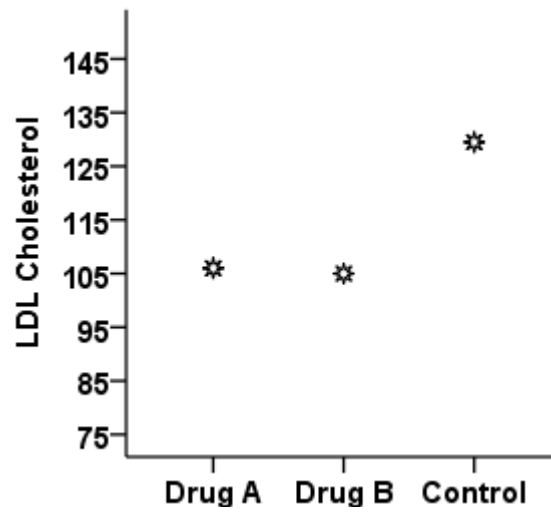
$$\psi = [(-.50)(\bar{X}_{\text{Drug.A}})] + [(-.50)(\bar{X}_{\text{Drug.B}})] + [(+1)(\bar{X}_{\text{Control}})] \quad (19)$$

When I tested the non-pairwise contrast above, I obtained the following SPSS table entitled ‘Contrast Tests’ (**Watch Video 7.21: [Non-Pairwise Comparisons in SPSS](#)**). It can be seen that the non-pairwise comparison was statistically significant. Thus, the combined Drug.A and Drug.B mean was statistically significantly different to the placebo group mean, $t = 2.27$, $p = .031$. As can be seen in Figure C7.6, the results suggest that taking a drug helped reduce LDL cholesterol, in comparison to placebo. Given that the sample sizes were equal, the Drug.A and Drug. B means could be averaged for the purposes of calculating Hedge’s g . Unequal sample sizes may require N -weighted averages.

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Contrast Tests						
	Contrast	Value of Contrast	Std. Error	t	df	Sig. (2-tailed)
LDL	Assume equal variances	1	24.0000	10.57725	27	.031
	Does not assume equal variances	1	24.0000	10.48272	18.436	.034

Figure C7.6. Plot of Means Treatment and LDL Cholesterol



Notes. 1 = Incongruent; 2 = Moderately incongruent; 3 = Congruent.

Practice Questions

The questions below are all very practical. They require you to read a summary of a research scenario (based on a genuine scientific publication). Then, after opening the data file, you need to test the specified hypothesis. Naturally, as these are ‘real-world’ research scenarios, you should concern yourself with the assumptions associated with the particular statistical analyses you decide to employ.

What analysis should you employ? Very good question. I have created a list of steps for you to follow to help you generate a statistical analytic strategy that should lead you to a defensible answer in virtually all cases. The steps are essentially a break-down of all of the key pieces of information I described in the Foundations section of this chapter (i.e., all material preceding the ‘Advanced’ section). You can watch and listen to me walk you through these importance steps (**Watch Video 7.22: [One-way Between Subjects ANOVA - Steps](#)**). Once you have reviewed the steps, have a go at answering the following questions. They are all based on scientific publications, so, in addition to practicing your statistical skills, I hope you learn some scientific “facts” along the way.

1: What’s the best font size for a mobile phone?

The mobile phone industry is massive. A mistake by a phone designer can cost millions. Consequently, it is important to get things right. For example, what font size should be used as default in a mobile phone? Darroch, Goodman, Brewster and Gray (2005) investigated that question. Specifically, they gave participants the same mobile phone to read a passage of text (about a paragraph’s worth). The dependent variable was time to completion (the participants read the passages allowed, so that a time keeper could keep time). The independent variable was font size and had four levels: 4, 6, 8, 14. I have simulated some data to correspond closely to the results reported by Darroch et al. (2005). Test the null hypothesis of equal time to completion means across the four groups (**Data File: [mobile_font_size](#)**) (**Watch Video 7.P1: [Games Howell in SPSS - Best Mobile Phone Font Size? \(Practice 1\)](#)**).

2. Are different racial groups more mindful than others?

Mindfulness is a psychological attribute broadly relevant to awareness of the environment, openness to new information, and different perspectives. Neuser (2010) collected data from the Five Facet Mindfulness Questionnaire from 349 adult participants. Neuser (2010) also collected demographic information, including race/ethnicity. Test the null hypothesis of equal mindfulness means across the six race/ethnicity groups. (**Data File: [races_mindfulness](#)**) (**Watch Video 7.P2: [Oneway Between-Subjects ANOVA – Non-Significant Effect \(Practice 2\)](#)**). What F -value did you get? What was the p -value associated with the ANOVA F -value? Was the null hypothesis rejected? What was the effect size associated with the ANOVA?

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3: Are high-action gamers more likely to take risks in life?

Some research suggests that playing high-action video games affects behavior. Examples of high-action video games include car racing games and first-person shooters. Bailey and West (2017) were interested to determine whether there was a difference in risky behaviors across individuals who do not play video-games, individuals who play a moderate amount of high-action video games (less than 20 hours per week), and a high amount of high-action video games (more than 20 hours per week). One of the key measures of risky behavior was based on the average amount of money wagered during an online game of blackjack. Test the null hypothesis of equal means of money wagered across the three groups during the blackjack game (**Data File: action_gamers_risk**) (**Watch Video 7.P3: [Oneway Between-Subjects ANOVA - Video Games \(Practice 3\)](#)**).

4: Do different summarization strategies lead to different levels of summarization competency?

Friend (2001) was interested to know whether three different types of summarization strategies were equally good at helping students summarize written material. The three summarization strategies were 'argument repetition', 'generalization', and 'self-reflection'. A total of 147 university students were assigned randomly to the three different summarization strategy groups. Each student was taught one of the three different strategies. Next, the university students were required to read 5 expository articles and summarize them. Finally, the quality of the summaries from each participant was evaluated quantitatively with the 'inclusion of important information' measure (0 to 100%). Test the null hypothesis that the three summarization strategies yielded equally good summarization mean values. (**Data File: summarization_strategies**) (**Watch Video 7.P4: [Oneway Between-Subjects ANOVA - Summarizing Strategies \(Practice 4\)](#)**).

5: Does Soft-drink consumption relate to aggressive behavior in children?

Suglia et al. (2013) were interested to determine whether there were differences in aggression levels between children who drink different amounts of soft-drink. The mothers of 199 children⁹ (5-years-old) rated the amount of aggression levels their children exhibited and the amount of soft-drinks their children consumed per day. Suglia et al. (2013) partitioned the soft-drink data into four groups: 0 soft-drink, 1 soft-drink, 2 soft-drinks, 3-soft-drinks, and 4 or more soft drinks. Test the null hypothesis that the mean aggression levels were equal across the five soft-drink groups (**Data File: sugar_aggression**) (**Watch Video 7.P5: [Oneway Between-Subjects ANOVA - Soft-Drink Consumption \(Practice 5\)](#)**).

6. Cramming versus spacing for an exam?

⁹ The sample size of the Suglia et al. (2013) study was actually 2929. I specified a sample size of 199 to make this more interesting from an educational perspective.

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Rohrer and Taylor (2007) were interested to learn what practice question completion strategy was best to employ with respect to performance on an exam. Specifically, Rohrer and Taylor (2007) were interested in the difference between ‘massing’ (completing all of the practice questions immediately after the lesson) versus ‘spacing’ (splitting the practice questions across time). To evaluate the question, presented the undergraduates ($N = 60$) with a mathematical problem and a method to solve the mathematical problem. The undergraduates were then given four practice questions. One third of the undergraduates completed all four practice questions immediately after the lesson (‘massers’); one third completed only two of the practice questions immediately after the lesson (‘light massers’); and one third completed two of the practice questions immediately after the lesson and the remaining two practice questions one week later (‘spacers’). The massers and light massers completed the test on week 2 and the spacers completed the same test on week 3 (so, everyone completed the same test within one week of their last practice question). I have simulated data to correspond closely to the results reported by Rohrer and Taylor (2007) (**Data File: massers_spacers**). For practice, test the null hypothesis of equal test performance means across all three groups (**Watch Video 7.P6: [Oneway Between-Subjects ANOVA - Cramming \(Practice 6\)](#)**).

Practice Questions: Advanced

7: Cramming versus spacing for an exam (non-pairwise)?

As a follow-on to the practice question above relevant to massers and spacers (**Data File: massers_spacers**), Conduct a non-pairwise contrast such that the two massers groups are ‘combined’ into the same mean and compared against the spacers (**Watch Video 7.P7: [Non Pairwise Comparison - Cramming \(Practice 7\)](#)**).

8: What is the number of hours of sleep sweet spot?

Ferrie et al. (2011) were interested in testing the hypothesis that differences in the need for sleep would have an effect on intelligence. They hypothesized that neither too little nor too much sleep was the most optimal level of sleep for cognitive functioning. Instead, somewhere in the middle was hypothesized to be the sweet spot. Test such a hypothesis with the data I simulated to correspond closely to the Ferrie et al. (2011) study. (**Data File: sleep_reasoning**) (**Watch Video 7.P8: [Nonlinear \(Quadratic\) Contrast Analysis - Reasoning \(Practice 8\)](#)**)

9: “Keep it classy, San Diego”

Some research suggests that brand exposure is good, but only up to a certain point. Eventually, prospective customers get bored or irritated with the brand. To evaluate this issue, Kirmani (1997) had participants flip through one of four slightly different magazines. The feature

that distinguished the magazines was the number of times a mineral ad appeared for a fake company. The participants were allocated randomly to four conditions (between-subjects design): magazine with two, three, five, and seven mineral water ads. After the participants read through the magazine, they were asked questions about the mineral water ad. Specifically, the participants rated the perceived quality of the mineral water on a 9-point scale (higher scores indicated better quality). I simulated some data to correspond closely to the results reported by Kirmani (1997) (**Data File: ad_frequency**) (**Watch: Ad Frequency and Quality**).

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